

| Question                                                                 |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                        |   | Marks available |     |     |       |       |      |
|--------------------------------------------------------------------------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------|-----|-----|-------|-------|------|
|                                                                          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                        |   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5                                                                        | (a) | (i)  |  | crop growth on fields increases                                                                                                                                                                                                                                                                                                                                                                        |   |                 |     | 2   | 2     |       |      |
|                                                                          |     |      |  | fertilisers run into waterways                                                                                                                                                                                                                                                                                                                                                                         | ✓ |                 |     |     |       |       |      |
|                                                                          |     |      |  | plant growth in rivers and lakes increases                                                                                                                                                                                                                                                                                                                                                             | ✓ |                 |     |     |       |       |      |
|                                                                          |     |      |  | aquatic animals do not have enough oxygen                                                                                                                                                                                                                                                                                                                                                              | ✓ |                 |     |     |       |       |      |
|                                                                          |     |      |  | farmers' profits increase                                                                                                                                                                                                                                                                                                                                                                              |   |                 |     |     |       |       |      |
| award (2) for <b>all</b> 3 correct<br>award (1) for <b>any</b> 2 correct |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                        |   |                 |     |     |       |       |      |
|                                                                          |     | (ii) |  | award (1) for statement that in general the larger the population, the more ammonia is used e.g. Asia has the largest population and the largest ammonia use / the largest population uses the most ammonia<br>'demand for food' is equivalent to population<br><br>award (1) for recognising Africa as the exception e.g. Africa has the second largest population but the fourth largest ammonia use |   |                 | 2   | 2   | 2     |       |      |
|                                                                          | (b) | (i)  |  | ammonium sulfate<br><br>do not accept ammonia sulfate                                                                                                                                                                                                                                                                                                                                                  |   | 1               |     | 1   |       |       |      |
|                                                                          |     | (ii) |  | plants use nitrogen to make sugar<br>plants use nitrogen to make water<br>plants use nitrogen to make oxygen<br>plants use nitrogen to make protein                                                                                                                                                                                                                                                    |   | 1               |     | 1   |       |       |      |